

Lohith Mothukuri

AI/ML Engineer | Machine Learning Engineer | Generative AI

(740) 844-3457 | lohith.mothu7@gmail.com | USA | [Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Medium](#) | [PortVia](#) | [DeshMate](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with **4 Years** of hands-on experience building and deploying production machine learning and Generative AI systems in financial services, currently at Visa. Develops transaction-risk, dispute-propensity, and anomaly-detection models (**XGBoost**, **PyTorch**) and Agentic **RAG** pipelines (**LangChain**, GPT-4) that serve 1M+ daily transactions. Skilled across the full ML lifecycle, including data pipelines, feature engineering, model training and fine-tuning, REST API deployment on AWS, CI/CD, and production monitoring. Holds an MS in Computer Science from Auburn University.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, SciPy), SQL (PostgreSQL, MySQL, Oracle), JSON

Machine Learning: XGBoost, Scikit-learn, Optuna, Isolation Forest, Logistic Regression, Clustering, Statistical Modeling

Deep Learning: PyTorch, TensorFlow, Keras, Neural Networks (ANN, RNN), Fine-Tuning, Reinforcement Learning (RLHF)

NLP & Generative AI: Hugging Face Transformers, spaCy, Sentence-Transformers, RAG, Agentic AI, Prompt Engineering, Vector Databases (FAISS, Pinecone, ChromaDB)

LLM & Agent Frameworks: LangChain, LangGraph, LlamaIndex, GPT-4, Claude, Mistral, Model Context Protocol (MCP)

MLOps & Deployment: FastAPI, MLflow, DVC, Docker, Prometheus, Evidently AI, Pydantic, Git, CI/CD

Cloud & Data Engineering: AWS (EC2, Lambda, S3), Apache Airflow, Databricks, DBT, Amazon Redshift

BI & Analytics: Power BI (DAX, Power Query, Data Modeling), Advanced Excel, Data Governance & Quality

PROFESSIONAL EXPERIENCE

Machine Learning Engineer – Gen AI

VISA, Atlanta, GA | Aug 2024 - Present

- Developed transaction dispute-propensity models (**XGBoost**, **Optuna**) with feature engineering and probability scoring, improving precision by 12% across 1M+ daily transactions.
- Built anomaly-detection pipelines (Isolation Forest, **PyTorch**) that identified irregular transaction behavior and reduced false positives by 15%, strengthening fraud-investigation workflows.
- Designed and deployed an Agentic **RAG** pipeline (**LangChain**, **LangGraph**, **FAISS**, GPT-4) with prompt engineering and context-window optimization, reducing policy-lookup time by 35%.
- Engineered AI-agent workflows using Model Context Protocol (**MCP**) and validated LLM outputs against structured JSON schemas with Pydantic, cutting malformed responses by 25%.
- Deployed models as **FastAPI** REST APIs on **AWS** Lambda, instrumented with **Prometheus** and **Streamlit** dashboards to monitor performance and model drift, sustaining sub-200ms p95 inference latency.
- Created **NLP** pipelines (**Hugging Face** Transformers, **spaCy**, Sentence-Transformers) for text classification and entity extraction, improving downstream financial-risk scoring inputs by 15%.
- Orchestrated experiment tracking, dataset versioning, and reproducibility (MLflow, DVC, Git) across 20+ experiments within Agile and CI/CD workflows.
- Collaborated with product, compliance, and engineering stakeholders to ship 5+ ML solutions from requirements to production, aligned with data-governance standards.

Data Scientist

American Airlines, Hyderabad, India | Dec 2022 - Jul 2023

- Built flight-cancellation classification models (XGBoost, Logistic Regression) on weather and crew data, achieving 87% accuracy and enabling proactive operational planning.
- Engineered features and preprocessing (NumPy, SciPy, Scikit-learn), improving dataset quality and reducing training variance by 18%.
- Automated data pipelines (Pandas, Apache Airflow, AWS S3) consolidating 5+ operational systems into clean, model-ready datasets.
- Analyzed boarding and delay data across 3+ hub airports (Python, SQL) to support on-time performance reporting.
- Developed Power BI dashboards (DAX, Amazon Redshift, DBT) tracking load-factor and fuel-usage KPIs across 3+ hubs for real-time operational visibility.

Senior Representative, Data Analyst (BFSI)

Concentrix, Hyderabad, India | Sep 2021 - Dec 2022

- Delivered quantitative analysis and reporting for 5+ enterprise BFSI clients across risk management, compliance monitoring, and operational decision-making.
- Migrated legacy queries to Oracle SQL and built Power Query ETL workflows, improving query performance by 25% and data availability by 35%.
- Designed 10+ interactive Power BI dashboards (DAX, data modeling), accelerating finance-stakeholder decisions.
- Performed exploratory data analysis and feature engineering (Python, Pandas, NumPy) to detect anomalies across 1M+ financial records, building domain foundations for fraud modeling.
- Automated reporting (Excel macros, Power BI Service), reducing manual effort by 40% and establishing data-quality and governance frameworks.

Process Associate, Real Estate Analytics

RealPage, Hyderabad, India | Jul 2021 - Sep 2021

- Cleaned and validated real-estate transaction datasets (Excel, SQL, data-quality checks), improving data consistency.
- Built 15+ recurring Excel and Google Sheets reports covering leasing and operational metrics, cutting manual report-preparation time by 30% for stakeholders.

PROJECTS

DeshMate – Cultural Roommate-Matching Platform - <https://www.deshmate.com>

- Built and launched a production roommate-finder marketplace for immigrant students and professionals across 36+ communities, with Google OAuth and verified-email auth, photo listings, map-based search, saved rooms, reviews, and an admin moderation dashboard with automated removal emails.
- Engineered a zero-framework stack for speed and cost (sub-second loads, near-zero infra spends) PostgreSQL with row level security, serverless email APIs, CDN edge deploys with CI smoke tests and uptime alerting, shipping daily production releases to real users.

Tech Stack:

JavaScript (ES6), HTML/CSS, Supabase (PostgreSQL, RLS, Auth, Storage), Vercel Serverless Functions, Google OAuth 2.0, Resend API, Leaflet.js, GitHub Actions, Cloudflare DNS

Porzolio - Portfolio Builder Web App - <https://porzolio.vercel.app/>

- Built and deployed a no-code portfolio builder where users drag, resize, and style blocks (text, links, images, video, audio, PDFs, resumes) on a free-form canvas and publish them instantly as a shareable public profile, with multi-page workspaces, starter templates, and an opt-in Explore directory.
- Engineered a custom vanilla-JavaScript canvas engine (selection, drag/resize, z-ordering, undo, rich-text toolbar) bridged into React, persisted each site as a single JSON block document in Supabase Postgres with autosave, secured multi-tenant data using Row Level Security and Supabase Auth (Google OAuth, magic link), sanitized user HTML to prevent stored XSS, and enriched links via Microlink, YouTube Embed, and GitHub REST APIs.

Tech Stack: Next.js 16, React 19, Supabase (Postgres, Auth, Storage), Vanilla JS, PostgreSQL RLS, Vercel

Customer Churn Prediction - ANN Binary Classification ([GitHub](#))

- Built a TensorFlow/Keras neural network for churn prediction with feature engineering and hyperparameter tuning. Stack: Python, TensorFlow, Keras, Scikit-learn.

IMDB Sentiment Analysis - RNN Text Classification ([GitHub](#))

- Developed a SimpleRNN (Keras) for binary sentiment classification with tokenization and embedding layers. Stack: Python, TensorFlow, Keras, NLP.

EDUCATION

Master of Science, Computer Science - Auburn University at Montgomery, Montgomery, AL | 2023 - 2025

Bachelor of Technology, Electronics & Communication Engineering - Anurag University, India | 2016 - 2020

COMMUNITY & THOUGHT LEADERSHIP

- Author AI/ML technical articles on [Medium](#) and share applied machine-learning insights with a growing professional audience on LinkedIn.